

SE LAB-1

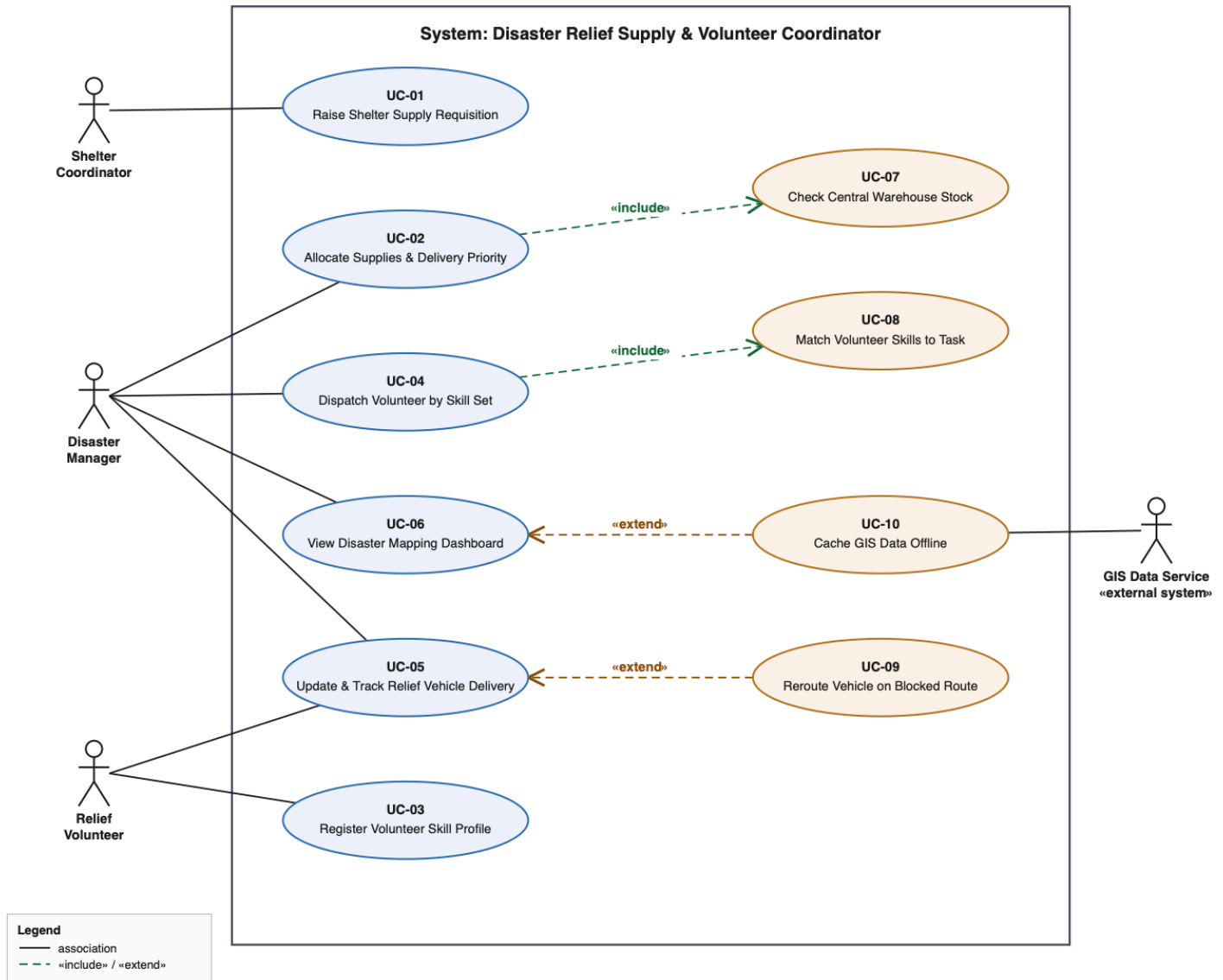
Name - PES1UG24AM915

Section - F

Requirments Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale	Traces to Use Case
FR-001	Functional	The system shall match incoming field shelter supply requisitions against available central warehouse supplies and allocate delivery priority.	High	Pass: A critical need (water / medical kit) is given the highest dispatch priority and is routed to the logistics manifest. Fail: A low-priority item is dispatched ahead of a water or medical need.	Core relief allocation logic; incorrect prioritisation directly costs lives at the shelters.	UC-02, UC-07
FR-002	Functional	The system shall allow a Shelter Coordinator to submit a supply requisition specifying item type (water, blankets, medical kits), quantity, shelter ID and severity level.	High	Pass: On submit, the requisition is stored with a unique ID and appears in the Disaster Manager's pending queue within 5 s. Fail: A requisition is accepted with item type, quantity, shelter ID or severity missing.	Requisitions are the entry point of the relief supply chain; incomplete records cannot be prioritised by FR-001.	UC-01
FR-003	Functional	The system shall allow a Relief Volunteer to register a skill profile (medical, driving, logistics, translation) together with an availability window.	High	Pass: A saved profile is retrievable and the volunteer appears in a skill search for every registered skill. Fail: A volunteer with no saved skill appears in the dispatch pool.	Verified skill data must exist before the skill-based dispatch in FR-004 can be trusted.	UC-03
FR-004	Functional	The system shall allow a Disaster Manager to dispatch volunteers to a shelter task by matching the task's required skill set against registered, available volunteer profiles.	High	Pass: Only volunteers whose skill matches and whose availability covers the task window are listed, and each dispatched volunteer is notified within 30 s. Fail: An unavailable or skill-mismatched volunteer is assigned.	Delivers the volunteer skill-set dispatching named in the problem context.	UC-04, UC-08
FR-005	Functional	The system shall track every relief vehicle delivery from dispatch to shelter handover, recording the states Dispatched, In Transit, Delivered and Failed with a timestamp for each transition.	Medium	Pass: Each logistics manifest shows its current state plus a timestamped history of all transitions. Fail: A manifest in the In Transit state has had no status update for more than 30 minutes.	Gives the control room accountability and visibility over relief vehicles in the field.	UC-05, UC-09
NFR-001	Nonfunctional (Performance & Security)	The disaster mapping dashboard must operate with low network bandwidth consumption and support offline GIS data caching.	High	Pass: Benchmarking under simulated peak load confirms the target latency and security standards - a map view transfers <= 500 KB and renders within 3 s, the last synchronised GIS layers still render with the network disabled, and the cache is stored encrypted. Fail: Any threshold is exceeded or cached data is stored unencrypted.	Disaster zones have degraded connectivity; the dashboard must stay usable offline and the cached map data must stay protected.	UC-06, UC-10
NFR-002	Nonfunctional (Reliability)	The system shall queue requisition and delivery-status updates entered while a device is offline and synchronise them automatically within 60 s of connectivity being restored, with no loss or duplication.	High	Pass: All 20 updates entered offline appear on the server within 60 s of reconnection, with no duplicates. Fail: Any update is lost or written twice.	Field devices lose connectivity frequently; silent data loss would corrupt the allocation picture built by FR-001.	UC-01, UC-05

Use Case Diagram



Use Case Flow

- Use Case ID / Name UC-02 – Allocate Supplies & Delivery Priority
- Primary Actor - Disaster Manager

2. Preconditions

1. The Disaster Manager is authenticated and holds the "Allocation" role.
2. At least one shelter supply requisition created by UC-01 is in the "Pending" state.
3. Central warehouse stock records are synchronised and readable by the system.
4. At least one relief vehicle is registered and marked available in the fleet register.

3. Postconditions

1. On success the requisition status is "Allocated" and a delivery priority is recorded against it.

2. A logistics manifest is generated and linked to exactly one assigned relief vehicle.
3. Central warehouse stock is decremented by the allocated quantity.
4. The assigned Relief Volunteer is notified and the allocation is written to the audit log.
5. On failure no stock is reserved; the requisition stays "Pending" or becomes "Escalated".

4. Main Success Scenario

1. Disaster Manager opens the Pending Requisition queue on the disaster mapping dashboard.
2. System lists every pending requisition with shelter ID, item type (water / blankets / medical kits), requested quantity and shelter severity level.
3. Disaster Manager selects one requisition to allocate.
4. System checks the requested items against central warehouse stock. — «include» UC-07 Check Central Warehouse Stock
5. System computes the delivery priority from item criticality (water and medical kits = Critical) and the shelter severity level.
6. System displays the proposed allocation, the computed priority, and the nearest available relief vehicle.
7. Disaster Manager confirms the allocation.
8. System reserves the stock, generates a logistics manifest, and assigns the manifest to the selected relief vehicle.
9. System notifies the assigned Relief Volunteer and sets the requisition status to "Allocated".
10. System writes an audit entry and displays the allocation confirmation.
11. Use case ends successfully.

5. Alternate Flow

- 4a. Insufficient Warehouse Stock — branches from step 4 of the main success scenario
 - 4a1. System detects that the available stock is less than the requested quantity and flags the shortfall on the requisition.
 - 4a2. System offers the Disaster Manager two options: allocate partially, or source the shortfall from an alternate warehouse.

- 4a3. Disaster Manager selects partial allocation. System allocates the available quantity and keeps the residual requisition in "Pending" with its original priority unchanged.
- 4a4. Flow resumes at step 5 of the main success scenario for the allocated quantity.
- 4a5. If no stock exists at any warehouse, the system marks the requisition "Escalated", notifies the procurement contact, and the use case ends without a logistics manifest.